



BITS Pilani
K K Birla Goa Campus

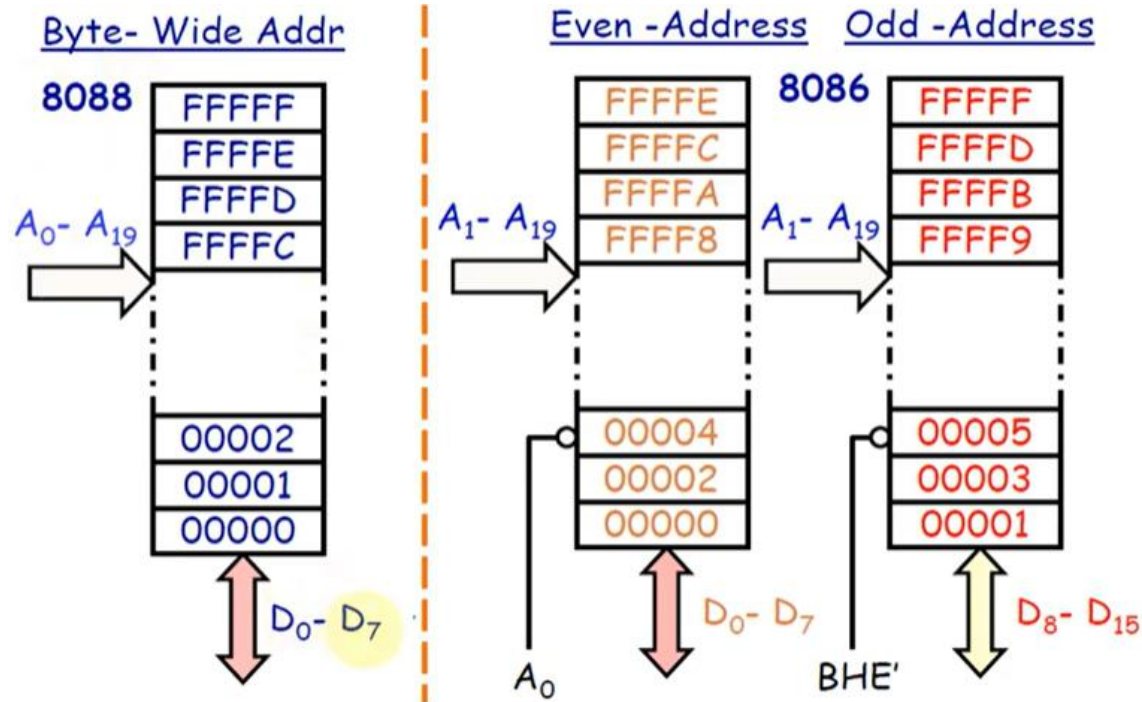
8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

Dr. Manideepa Mukherjee
Department of CS & IS



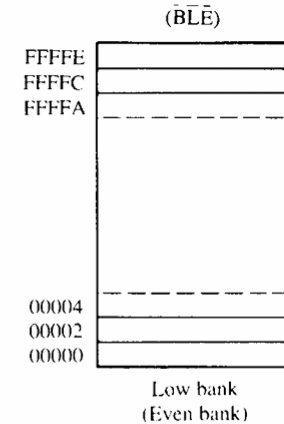
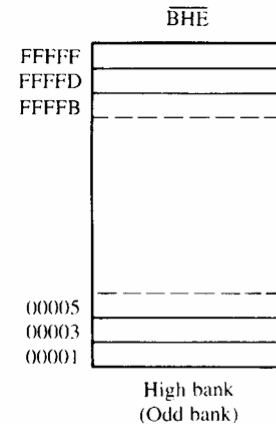
Memory Interfacing with 8086



Memory Interfacing with 8086

$\overline{BHE}$	A_0	Indication
0	0	Whole word (2 bytes)
0	1	Upper byte from or to odd address.
1	0	Lower byte from or to even address
1	1	None

BHE	$\overline{BLE}$	Function
0	0	Both banks enabled for a 16-bit transfer
0	1	High bank enabled for an 8-bit transfer
1	0	Low bank enabled for an 8-bit transfer
1	1	No bank enabled



Note: A_0 is labeled $\overline{BLE}$ (Bus low enable) on the 80386SX.

Memory Interfacing: 8086

Address	Data Type	BHE'	A ₀	Bus Cycles	Data lines used
00000	Byte	1	0	one	D ₀ -D ₇
00000	Word	0	0	one	D ₀ -D ₁₅
00001	Byte	0	1	one	D ₈ -D ₁₅
00001	Word	0	1	first	D ₈ -D ₁₅
		1	0	second	D ₀ -D ₇

Memory Interfacing: 8086

If memory chips available are only 2KB each. Do the interfacing for 8086

Memory Requirements:

2716 –size 2K

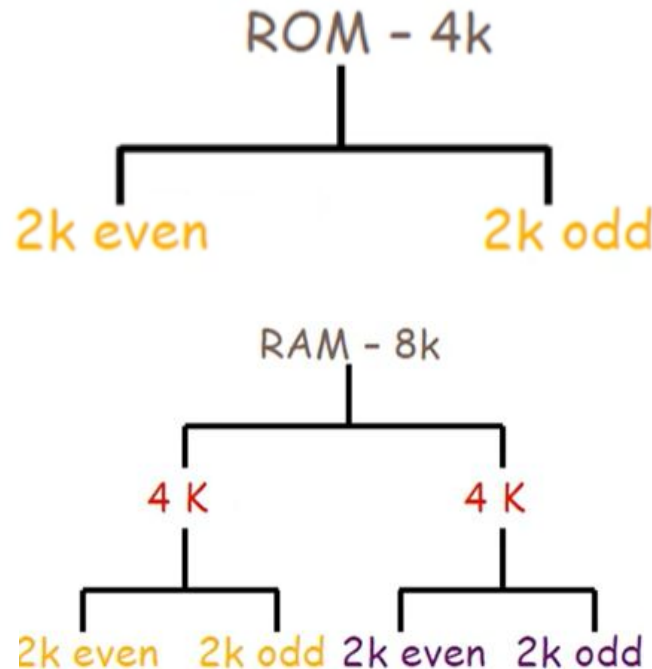
ROM –4k

Number of 2716 –**2**

6116 –size 2k

RAM –8k

Number of 6116 - **4**



Memory Interfacing: 8086

If memory chips available are only 2KB each. Do the interfacing for 8086

Ex: Interface 4K 2716 (ROM) starting at 00000H 8K 6116 (SRAM) starting at 08000

ROM₁ **00000_H - 00FFF_H**

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1

Memory Interfacing: 8086

RAM₁

08000_H - 08FFF_H

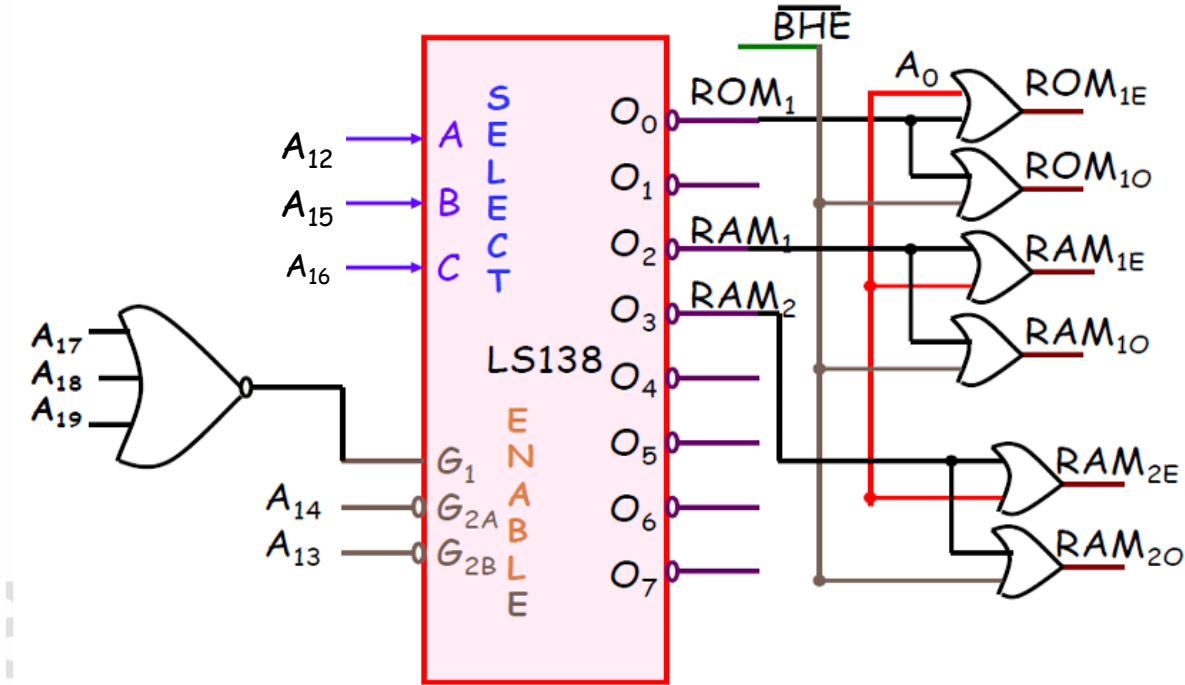
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1

RAM₂

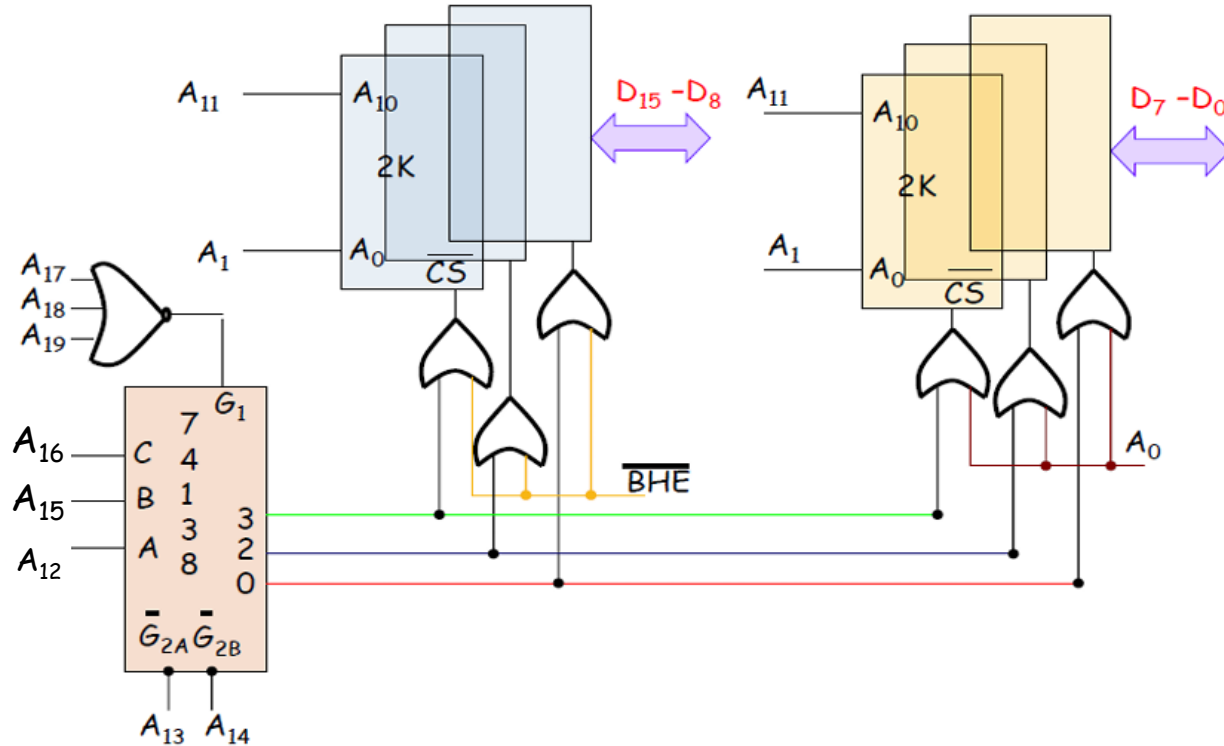
09000_H - 09FFF_H

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1

Memory Interfacing: 8086



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Decode 1 MB memory using
16 bit bus

